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The Effect of Mass on a Damped Pendulum

004180-0075

Research question

How does the mass added to a damped pendulum bob (0, 10, 20, 30, 40, 50, 60, 70g) affect its horizontal amplitude from equilibrium after 10 periods?

Background Information

Simple Harmonic Motion (SHM) is defined as oscillatory motion in which the restoring force is proportional and opposite to displacement, and acts towards the equilibrium position. The period is constant, and the displacement time graphs can be modelled by sinusoidal graphs.

A classic example of SHM is the simple pendulum, consisting of a small bob and a massless string. In a vacuum without air resistance, the pendulum will continue to oscillate indefinitely with a constant amplitude.

For small angular displacements, the small-angle approximation can be applied, where $sin\theta \approx \theta$ and θ is less than about 15° (The Simple Pendulum, 2025). Under this condition, the motion of a simple pendulum can be approximated as SHM, and the period T is given by Eq. 1.

$$T = 2\pi\sqrt{\frac{L}{g}} \tag{1}$$

Where L is the length of the pendulum, and g is the acceleration due to gravity 9.81 ms^{-2}.
This equation shows that the period is independent of the mass of the bob.

The equation for the acceleration of SHM is given as:

$$a = -\omega^2 x \tag{2}$$

Where ω is the angular velocity and $\omega^2 = \frac{g}{L}$. x is the displacement from equilibrium.

However, in air, resistive forces act on the pendulum, causing it to gradually lose mechanical energy over time. This phenomenon, known as damping, results in the amplitude of a pendulum continuously decreasing while its period and frequency remain constant (Mohazzabi et al., 2017).

The forces acting on the pendulum include tension in the string and the gravitational force mg, which can be resolved into two components.

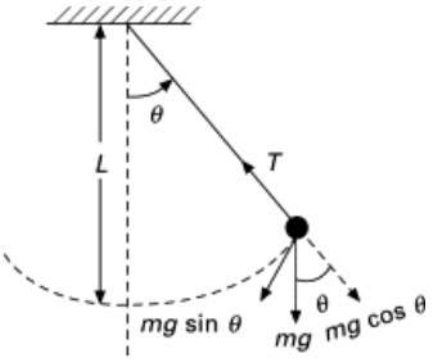


Figure 1: Forces acting on a simple pendulum
Source: Rajasekaran, 2009

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As seen in Figure 1, $mgcos\theta$ acts along the string and $mgsin\theta$ acts tangential to the arc of motion, and is responsible for the restoring force (Rajasekaran, 2009). Additionally, air resistance (drag) acts opposite to the direction of motion, doing work against the system. This results in the continuous loss of mechanical energy in the form of kinetic and potential energy, transformed into thermal energy as a result of the frictional interactions between the pendulum bob and air molecules.

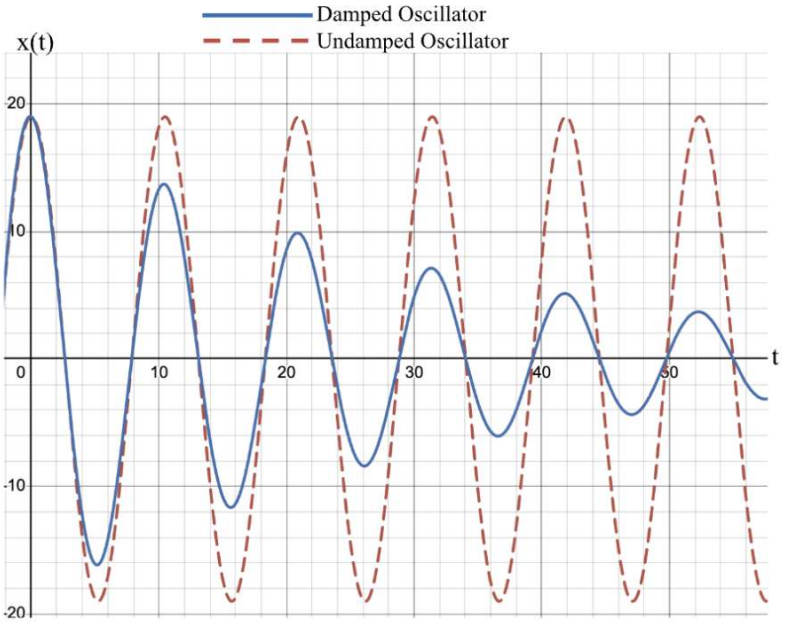


Figure 2: Amplitude of an undamped vs. damped oscillator over time

The damping force, referred to as drag, is typically proportional to the velocity of the object for low-speed motion in air:

$$F_{drag} = -bv \tag{3}$$

Where b is a damping coefficient that depends on the properties of the medium (air) and the geometric properties of the ball (Mobolaji, 2017).

For a spherical object moving through a viscous medium at low speeds, the drag force can be approximated by Stokes' Law (Cartwright, 2020):

$$F_{drag} = 6\pi\eta rv \tag{4}$$

Therefore, for this experiment, this can be written as:

$$F_{drag} = -bv \quad \text{where} \quad b = 6\pi\eta r$$

In the presence of air resistance, the net force includes both the restoring force due to gravity and the damping force due to drag:

$$F_{net} = ma \tag{5}$$

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$$F_{net} = F_{restoring} + F_{drag}$$

$$F_{restoring} + F_{drag} = -m\omega^2 x - bv$$

$$ma = -m\omega^2 x - bv$$

This can be rearranged as:

$$ma + bv + m\omega^2 x = 0 \tag{6}$$

To derive an equation that models x(t), we use the relationship that v is the first derivative of displacement with respect to time, and acceleration is the derivative of velocity. Therefore, by differentiating x(t) with respect to time, we obtain expressions for velocity and acceleration, which can be substituted into Eq. 6.

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + m\,\omega^2 x = 0$$

Dividing through by m, we obtain:

$$\frac{d^2x}{dt^2} + \frac{b}{m}\left(\frac{dx}{dt}\right) + \omega^2 x = 0$$

This is a second-order differential equation, with the solution: (The Damped, 2022)

$$x = -\frac{b}{2m} \pm \frac{1}{2}\sqrt{\frac{b}{2m}^2 - 4\omega^2}$$

The general rule for a second-order differential equation states that if the discriminant is negative, the complex roots of the equation give the general solution (Second Order, 2025)

$$x = e^{-\frac{b}{2m}t}(A_o cos(\omega t) + iDsin(\omega t))$$

After 10 periods, $iDsin(\omega t)$ it can be omitted because $sin(20\pi) = 0$, as shown below:

$$x = e^{-\frac{b}{2m}t}(A_o cos(\frac{2\pi}{T}10T) + iDsin(\frac{2\pi}{T}10T))$$

$$x = e^{-\frac{b}{2m}t}(A_o cos(20\pi) + iDsin(20\pi))$$

$$x = e^{-\frac{b}{2m}t} A_o cos(20\pi)$$

To model the underdamping of the pendulum, which continues to oscillate, the discriminant $\sqrt{\frac{b}{2m}^2 - 4\omega^2}$ must be < 0 (Stuart, 2025).

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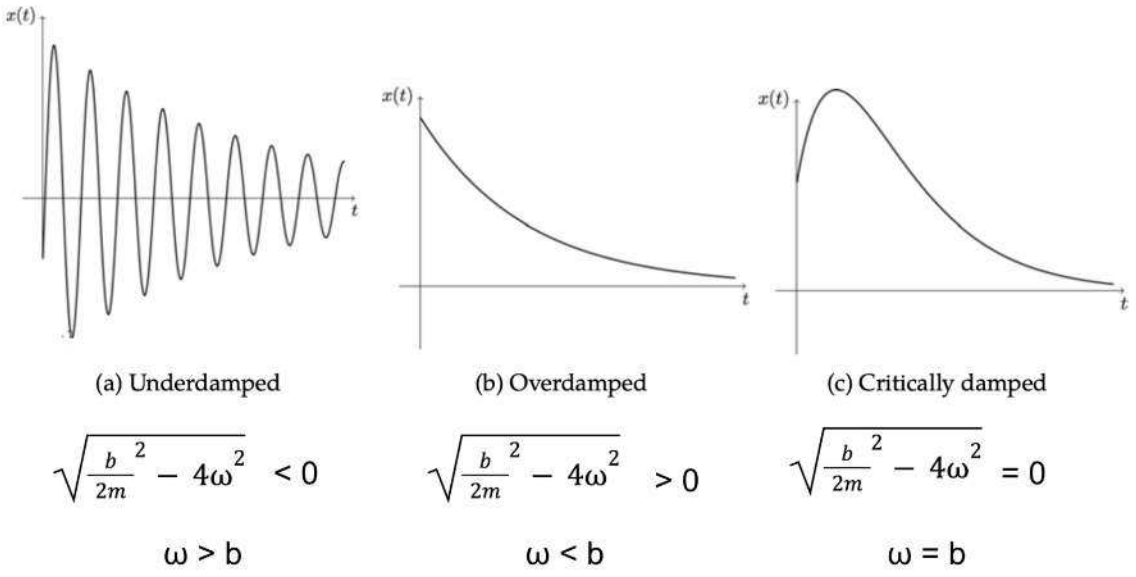


Figure 3: Different types of damped oscillations

Thus, the motion of a damped pendulum can be modelled by the equation (Damped Oscillations, 2022)

$$x(t) = A_o e^{-\frac{b}{2m}t} cos(\omega t) \tag{7}$$

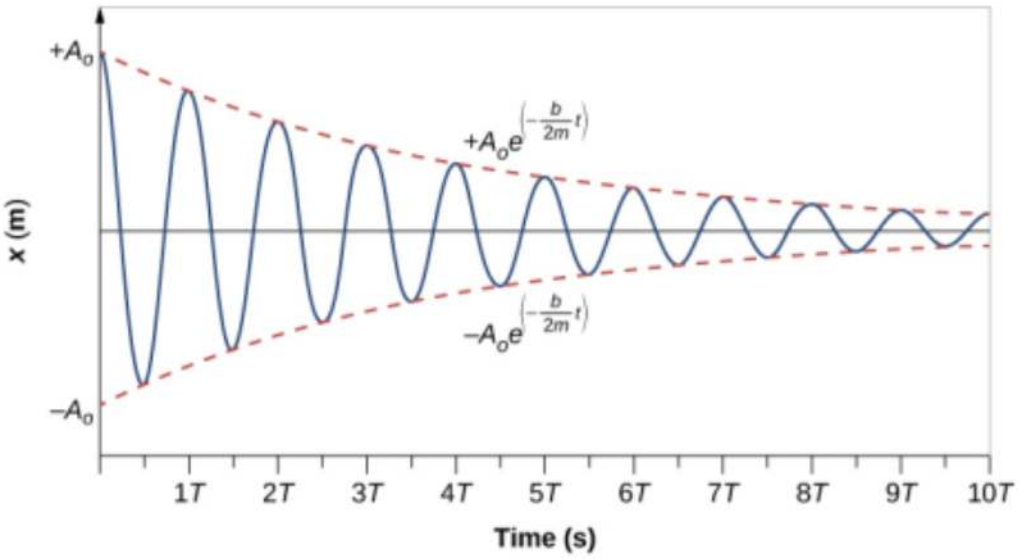


Figure 4: Variation of maximum amplitude with time of a damped oscillator.
Source: Damped Oscillations, Physics LibreTexts 2022

Therefore, if we are only interested in the maximum amplitude of the oscillation at a given time, the motion of the damped pendulum can be modeled using the equation:

$$x(t) = A_o e^{-\frac{b}{2m}t} \tag{8}$$

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Taking the natural logarithm of both sides linearizes the relationship:

$$ln\,x = lnA_o - \frac{b}{2m}t$$

Since maximum amplitude is measured after 10 full oscillations $t = 10T$, this becomes:

$$ln\,x = lnA_o - \frac{5bT}{m} \quad (9)$$

This equation shows that $ln\,x \propto -\frac{1}{m}$, meaning that as the mass increases, the rate of amplitude decay decreases.

If a graph of $ln\,x$ is plotted against $\frac{1}{m}$, a linear relationship is expected to be observed. The gradient of this line is predicted to be $-5bT$, as well as have a y-intercept of lnA_o

Experiment Methodology & Design

Variables

Independent Variable	Dependent Variable
Mass added to a pendulum bob 0, 10, 20, 30, 40, 50, 60, 70g This range was determined through preliminary trials. For masses greater than 70g, the difference in amplitude after $10T$ was very small.	**Horizontal amplitude** from equilibrium after 10 periods. Measured by LoggerPro video analysis. This was chosen because preliminary trials showed that there was a significant decrease in maximum amplitude after 10 periods.

Controlled Variable	Why is it controlled?	How is it controlled?
Period	Since the maximum amplitude is measured after a fixed number of oscillations (e.g., 10 periods), the period must remain constant to ensure a fair comparison.	Max amplitude is consistently measured after 10 full oscillations in all trials.
Initial/maximum amplitude	The initial amplitude affects the energy of the system. To isolate the effect of mass of damping, the initial displacement must be the same for all trials.	The pendulum is always released from the same height (87.0 cm on a meter ruler), using a reference on a bench, with no push applied.
Shape, size, texture, and type of pendulum capsule ball.	The volume, surface area, and texture of the capsule ball all affect the amount of air resistance and friction it experiences.	The same plastic capsule ball is used for all trials. It was chosen because it would maintain size, volume, and surface area while being able to insert different masses inside.

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Mass attachment method	Shifting of weights inside the capsule would affect the center of mass and cause variation in motion.	Small amounts of Blu-Tack were used to secure all internal masses to the capsule's interior so they do not move during oscillation.
Length and type of string	The period of a pendulum depends on its length. $T = 2\pi\sqrt{\frac{l}{g}}$ An elastic string may stretch when under heavier loads, thus changing the length of the string. Therefore, an inelastic, inextensible string was used.	An inelastic, inextensible string is used. One end of the string was binder clipped to a meter stick, and the other end tied to slits in the pendulum bob. The length of the string used was relatively long (65.7cm) to stick to the small-angle approximation while having a large enough displacement, hence reducing fractional uncertainty.
Position of the video recording phone	To minimize and standardize parallax error and maintain consistent framing and scaling in video analysis, the phone position and type must not change between trials.	The phone was secured using a phone holder and placed directly opposite the motion plane. Same phone (iPhone 15, 1x zoom) used throughout.
Equilibrium position	The equilibrium point defines the zero reference for measuring amplitude. An inconsistent equilibrium reference would distort displacement measurements.	A black reference line is drawn on the graph paper at the equilibrium position (29.4cm on the meter ruler), aligned with the center of the bob at rest.
Where the position of the capsule is read	Measurement must always refer to the same point on the capsule to avoid variation in displacement data.	The center of the capsule is used as the reference point for all position readings.
Environmental Conditions	Drafts, fans, or air currents could affect damping and lead to inconsistent results.	All trials are conducted in the same indoor space with windows closed and air conditioning maintained at a constant temperature.

Apparatus

Apparatus Name	Precision (where applicable)	Quantity
Ohnaus Navigator NV622 2 d.p. Electronic scale	± 0.01 g	1
Retort Stand		2
Clamp		2
Meter Ruler	± 0.05cm	2
Inelastic string/thread		1
Capsule surprise ball		1

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Medium-sized binder clip		1
Large whiteboard		1
A4 graph paper		6
Clear tape roll		1
Benches/seats		3
Phone Stand		1
Recording Device (iPhone 15)		1
50g weight		1
20g weight		2
5g weight		2

Methodology

Preliminary trials

The maximum mass of small weights that could fit inside the empty capsule is 250g, however, preliminary trials determined that after 70g, there was not a noticeable difference in maximum amplitude as mass increased. Therefore, a range of masses added was chosen to be 0, 10, 20, 30, 40, 50, 60, and 70 grams.

A noticeable effect was observed after 5T for smaller masses, but not for larger masses. Therefore, 10T was chosen because after 10T, there is a significant difference in the reduction of horizontal amplitude for the large masses.

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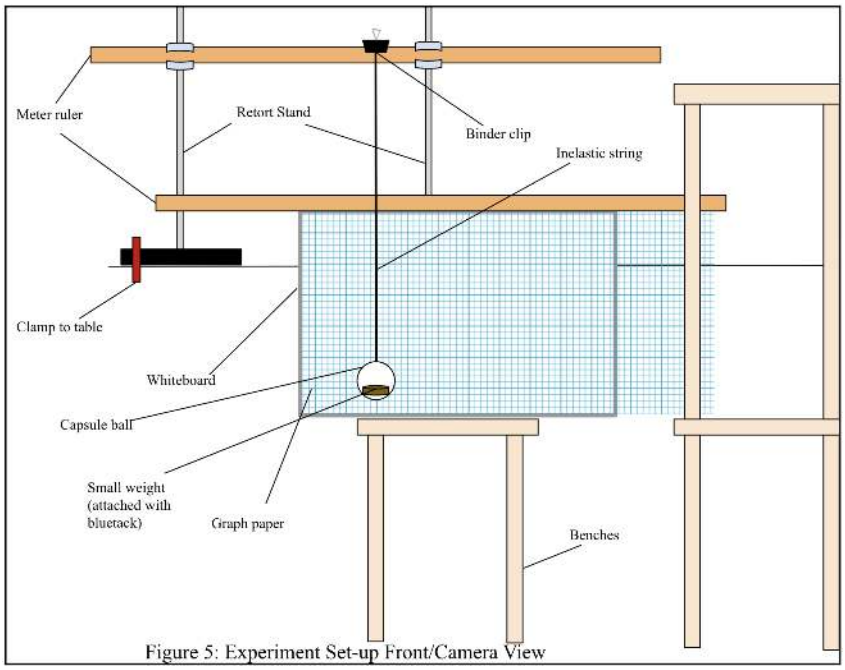


Figure 5: Experiment Set-up Front/Camera View

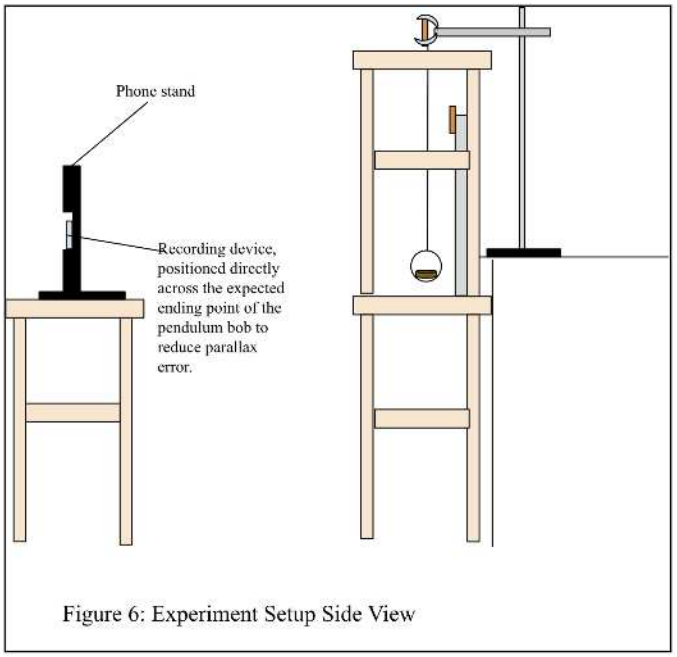


Figure 6: Experiment Setup Side View



Figure 7: Photograph of experiment set-up

1. Small weights (0, 10, 20, 30, 40, 50, 60, 70g) were secured inside a plastic capsule using Blu Tack to prevent internal movement, ensuring a fixed center of mass and preventing any inconsistencies. The total added mass was measured using a 2-decimal-place electronic scale.
2. Two narrow slits were cut into the capsule ball using a penknife, through which an inextensible, inelastic string was threaded and secured with multiple knots. This ensured the string remained fixed in length and allowed easy mass changes. Tape was avoided to prevent length variation or slippage.
3. A meter ruler was clamped horizontally between two vertical retort stands, each secured to the lab bench with table clamps. Two stands were used instead of one to avoid bending under load, ensuring accurate amplitude measurements.
4. The pendulum was suspended using a binder clip clipped to the meter ruler, between the two retort stands. A binder clip was selected to prevent movement at the suspension point,

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ensuring accurate string length. Tying the string to the ruler would allow the pivot point to oscillate slightly with the pendulum, introducing systematic error.

5. The pendulum length was measured using a meter stick from suspension point to the edge of the capsule where the string was tied. The length remained constant throughout all trials and was chosen to be relatively long (57.6cm). This is to satisfy the small-angle approximation, but also to have a large enough maximum displacement from equilibrium that can be measured with a lower percentage uncertainty.
6. A whiteboard was placed behind the pendulum, with graph paper affixed to it to improve amplitude measurement accuracy. A black horizontal line was drawn at the equilibrium position (29.4 cm) to serve as a reference point during LoggerPro video analysis.
7. A smartphone on a stable tripod was positioned perpendicular to the oscillation plane and aligned with the expected ending position, not the starting position, to minimize parallax error. The camera remained fixed throughout to standardize video measurements.
8. Video recording was started with a constant zoom level (1x).
9. The pendulum was released from a consistent point (87.0 cm) using identical bench legs as a reference. This ensured constant maximum amplitude and initial potential energy. The ball was dropped without any additional force to avoid introducing extra kinetic energy.
10. Each trial was recorded for at least 10 full oscillations.
11. Videos were transferred to a laptop and imported into LoggerPro. The origin was set at the equilibrium position (29.4), thus making the maximum starting amplitude $87.0 - 29.4 = 57.6$ cm. Points were placed at the approximate center of the capsule: once at the starting amplitude and again after 10 full oscillations. Values of $10T$ were also recorded through LoggerPro.
12. Steps 1-11 were repeated for each added mass of 0, 10, 20, 30, 40, 50, 60, 70 grams.

Data Analysis

Raw Data

Raw data of the measurements for x, the horizontal displacement from equilibrium, were recorded after 10 oscillations. Values for the independent variable, mass that was added to the capsule bob, were also recorded.

Mass added to the capsule pendulum bob (g) ± 0.01 g	Horizontal amplitude of the center of the pendulum bob after 10T (cm)						
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7
0.00	19.07	18.48	18.64	18.62	18.83	18.76	18.48
10.69	32.56	33.77	32.70	32.99	33.59	33.10	32.66
20.25	39.07	40.58	40.77	39.52	39.16	39.89	39.39
30.74	42.99	43.42	44.28	42.51	43.49	43.57	42.67
40.72	46.56	45.23	45.01	46.53	45.98	45.45	46.46

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52.58	48.03	49.10	48.41	48.68	47.80	46.80	49.00
60.98	48.44	47.80	48.70	48.93	49.10	48.86	49.95
70.31	48.02	49.43	50.59	48.31	47.93	47.01	48.62

Table 1: Raw data of mass added and horizontal amplitude after 10 periods.

Since this raw data is collected through LoggerPro, there is greater error from trial to trial than due to the measuring instrument. Thus, random error from the readings is considered through $\frac{max-min}{2}$ the natural logarithm of horizontal amplitude. Example calculations are shown in the Uncertainty Propagation section.

The table below shows raw data values for the time for 10 oscillations measured using LoggerPro. This was used to obtain an average value of T using an average of all values recorded.

Mass added to the capsule pendulum bob (g) ± 0.01 g	10 periods (s) ± 0.002 s							Average 10T (s)
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7	
0.00	17.072	17.138	17.098	17.105	17.098	17.098	17.132	17.106
10.69	17.402	17.438	17.435	17.442	17.402	17.440	17.432	17.427
20.25	17.535	17.533	17.540	17.535	17.538	17.598	17.570	17.550
30.74	17.598	17.635	17.603	17.607	17.640	17.640	17.603	17.618
40.72	17.933	19.180	18.075	17.707	16.040	17.673	17.672	17.754
52.58	18.100	18.008	18.100	17.765	17.705	17.668	17.665	17.859
60.98	17.740	17.708	17.770	17.742	17.770	17.615	17.772	17.731
70.31	17.675	17.865	17.780	17.735	17.35	17.742	17.770	17.702

Table 2: Time period of 10 oscillations for each trial, measured using LoggerPro

Mass of empty capsule = 6.04g ± 0.01 g
Length of string = 67.5 cm ± 0.1 cm
Diameter of ball = 4.97 cm ± 0.02 cm

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Processed Data

The natural logarithm of x (in meters) was calculated and averaged over seven trials for each value of the independent variable. The total mass was determined by adding the added mass to the capsule mass and then calculating $\frac{1}{m}$.

Example Calculations

Mass of pendulum bob: $6.04 + 10.69 = 16.73g = 0.01673$ kg

$\frac{1}{mass\ of\ pendulum\ bob} = \frac{1}{0.01673} \approx 59.77\ \text{kg}^{-1}$

$ln(\frac{19.07cm}{100}) = ln(0.1907m) = -1.657$

Average ln(x) $= \frac{-1.657-1.688-1.680-1.681-1.670-1.673-1.688}{7} = -1.676 \approx -1.68$

Uncertainty Propagation

Example Calculations

The mass added had an uncertainty of ± 0.01, mass of the empty capsule had ± 0.01. Adding together means that uncertainty adds together. So that equals a total of ± 0.02g.
Then, in the calculation of $\frac{1}{m}$ there is a division, and the fractional uncertainty is added together, described by the formula:

$\frac{\Delta m_{reciprocal}}{m_{reciprocal}} = \frac{\Delta m}{m}$

$\Delta m_{reciprocal} = m_{reciprocal} \times \frac{\Delta m}{m}$

$\Delta m_{reciprocal} = 165.56 \times \frac{0.02}{6.04} = 0.5482 \approx 0.5$

Uncertainty of ln(x) was determined by using $\frac{max-min}{2}$ ln(x) values for all trials.

$\frac{(-1.657)-(-1.688)}{2} = 0.0155 \approx 0.02$

Processed Data Table with Uncertainties

$ln(x)$	**Absolute uncertainty of** $ln(x)$	$\frac{1}{m}$ **(kg^{-1})**	**Absolute uncertainty of** $\frac{1}{m}$
-1.68	0.02	165.6	0.5
-1.11	0.02	59.77	0.07
-0.92	0.02	38.04	0.03

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-0.84	0.02	27.19	0.01
-0.78	0.02	21.39	0.009
-0.73	0.02	17.06	0.006
-0.72	0.02	14.92	0.004
-0.72	0.04	13.10	0.003

Graph & Conclusion

To investigate the relationship $ln\ x = lnA_o - \frac{5bT}{m}$, a graph of lnx against $\frac{1}{m}$ was plotted.

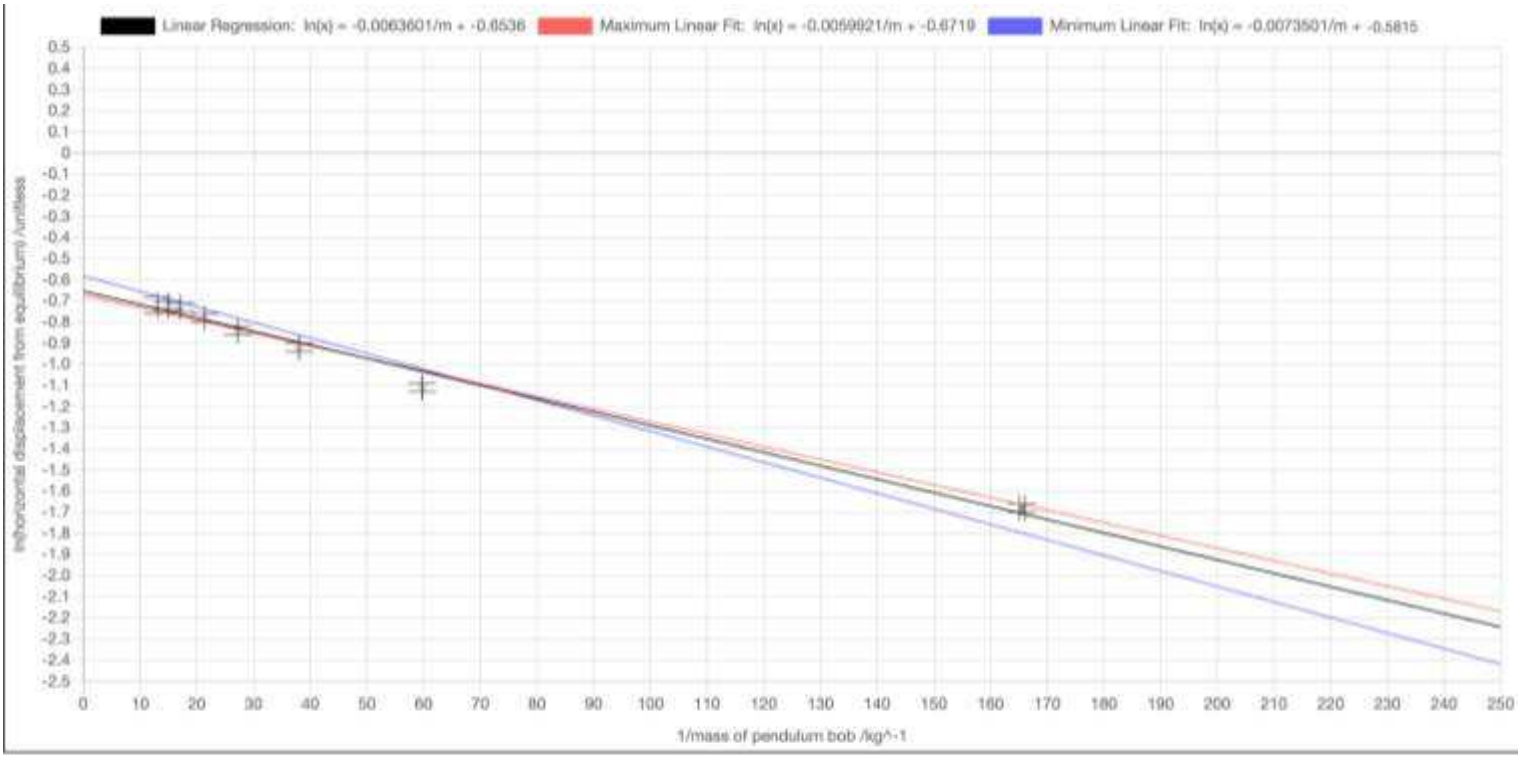


Figure 8: Natural logarithm of the horizontal displacement from equilibrium (lnx) against the reciprocal of the pendulum bob's mass (1/m)

Expected slope and y-intercept

At room temperature (20°C), air has a coefficient of viscosity of 1.8×10^{-5} Pa·s (Bao, 2000)

The radius of the capsule was measured as $\frac{4.97}{2} = 2.485$ cm

As can be seen in the Raw Data, different values for 10T were measured for each trial, but the average of all of them is 10T = 17.593 seconds, therefore:

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$T = \frac{17.593}{10} = 1.7593$ s

The expected slope is:

$-5(6\pi\eta r)T$

$= -5(6\pi)(1.8 \times 10^{-5})(2.485)(1.7593)$

$= -0.0074$ kg

As the max horizontal amplitude was measured at 57.6 cm for all trials, the expected y-intercept is:

$lnA_o = ln(\frac{57.6}{100}) = -0.552$

Experimental slope and y-intercept

The slope of the best-fit line is – 0.0064, which is less than the expected value of – 0.0074.

However, the expected slope, – 0.0074, is not within the range between the maximum slope and the minimum slope of – 0.007350. This indicates that the deviation error cannot be fully accounted for by random error, and there is the presence of a systematic error.

The y-intercept of the best fit line is -0.654, which is less than the expected value of – 0.552.

Once again, the expected y-intercept – 0.552 is not within the range of the maximum y-intercept and the minimum y-intercept. This means that the error cannot be accounted for by random error and is largely due to systematic error.

Although the slope and y-intercept deviate from expected values, this alone does not confirm that the relationship is non-linear. What more strongly suggests a non-linear relationship is the shape of the plotted data: the points curve away from the regression line, especially towards the right side of the graph. The point corresponding to $\frac{1}{m} = 165.6$ shows a noticeable deviation from the linear trend. Additionally, the regression line fails to pass through five of the data points, which further supports the observation that the data may not follow a strictly linear pattern.

This is largely due to a systematic error in the experiment when additional masses are added to the pendulum.

Figure 9 is a close-up of the left side of the graph. It shows another deviation from the expected linear trend.

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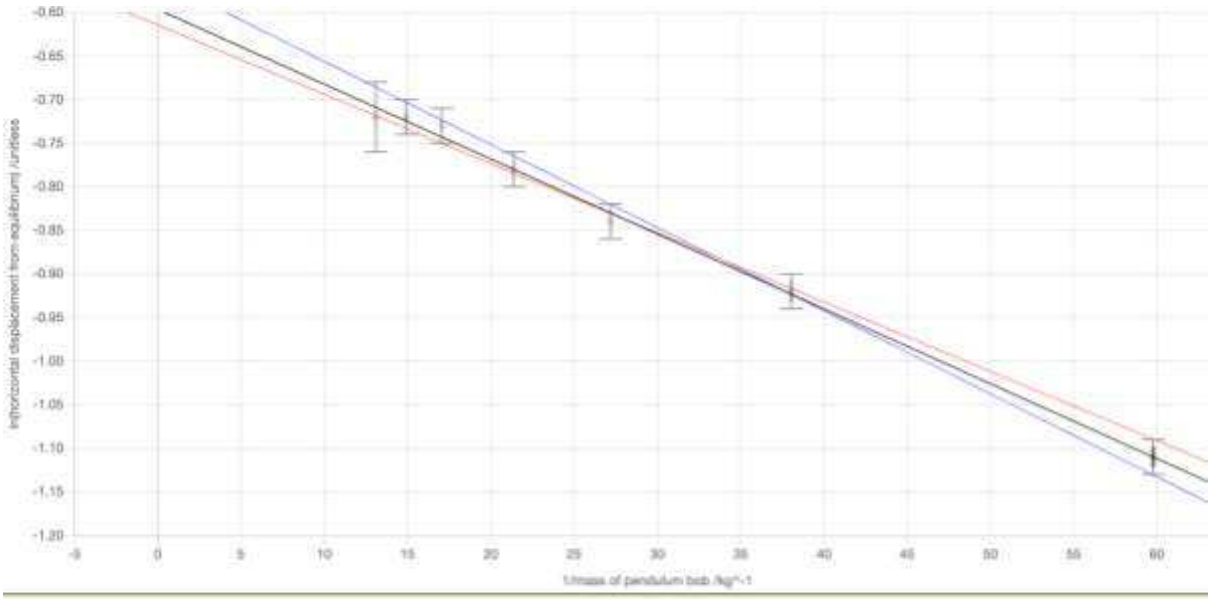


Figure 9: Close up graph of ln(x) against 1/m for smaller values of 1/m

The left part of the graph, where masses are larger (hence 1/m is smaller), shows a slight plateau that occurs as $\frac{1}{m}$ gets smaller and smaller. The values for ln(x) approach a constant value of -0.72. This could be because damping forces such as air resistance increase non-linearly with speed. As more mass is added, the effect of air resistance is reduced or becomes insignificant. Hence, the pendulum becomes harder to slow down, so it loses less amplitude with each swing. Therefore, beyond a certain point, adding more mass doesn't significantly affect the amplitude after 10T (Peters, 2003).

Evaluation

To assess the reliability of this investigation, key sources of random and systematic errors were identified. The table below outlines each error, its impact on the relationship that is observed, and suggested improvements to improve result validity.

Type of error	Error	Impact of error	Suggested improvement
Systematic	The mass was not centered on the pendulum.	The difference between the right-most point and the rest of the posts could be explained by a shift in the center of gravity of the pendulum when additional mass is added. The model assumes that the period of oscillation T stays constant. However, in practice, T depends on the effective length of the pendulum, measured from the pivot to the center of mass. Adding weights to the bottom lowers the center of mass and increases the effective length. As a result, T also increases. Moreover, as small weights are	Secure the mass at the center of the capsule each time using adhesive putty/Blu Tack or foam pads. Another option might be to use pendulum bobs of identical shape and surface finish but made from metals of different densities.

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		stacked on top of each other, the center of mass rises. Shortening the length and decreasing the period. This explains why the measured value of $10T$ increases initially, then increases more slowly. Additionally, shifting the center of gravity introduces torque, causing the pendulum to swing in out-of-plane motion rather than a 2D arc. This out-of-plane motion cannot be observed through the video analysis and would affect the accuracy of amplitude measurements.	
Systematic	The ruler was slanted or not leveled during measurement.	If the ruler is slanted towards the right (right below left), it would overestimate the reading of the amplitude. There is a systematic increase in all amplitude readings, while the actual amplitude is smaller, skewing the data.	Ensure that the ruler is perfectly horizontal using a spirit level.
Random	Manual release of the capsule is not uniform/consistent.	Uneven release introduces variation in the initial amplitude as well as amplitude after 10. If there is a slight push, then mass isn't the only factor that affects the damping anymore. Hand friction during release may even introduce a drag force, which increases the effects of damping. Sometimes the initial amplitude is not always released at 57.6cm. This can lead to larger or smaller amplitudes, leading to inconsistency.	Using a mechanical release system, such as an electromagnet or clamp. In an electromagnet, an electric current creates a magnetic field to hold the pendulum; switching off the current instantly drops it without extra force.
Random	Inconsistent scaling in Logger Pro. Each video was set up individually, leading to potential variation in the scale. (setting scale, origin, and axis differently for each video)	Inconsistent scaling affects the measured displacement after 10T, resulting in variability across the 7 trials. This reduces the precision of the data and causes larger error bars.	Record one long video for all trials, then the Logger Pro scaling would only have to be set up once. This keeps it consistent and reduces random error.

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